

WEATHER STATION Using STM32 (Blue Pill) And ThingSpeak IoT

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This simple weather station detects environmental parameters such as temperature, humidity, and soil moisture. The temperature and humidity sensor DHT11 used measures the atmospheric temperature while the soil moisture sensor measures moisture in the earth. Block diagram of the project is shown in Fig. 1.

Circuit for regulated power supply required for the project is shown in Fig. 2. It is built around 12V-0-12V, 750mA step-down

centre-tapped transformer, full-wave rectifier comprising two 1N4007 diodes (D1 and D2), 5V regulator 7805 (IC1), and two electrolytic capacitors: 1000 μ F, 25V (C1) and 100 μ F, 25V (C2).

The main circuit, shown in Fig. 3, is built around soil moisture sensor module DHT11, STM32F103 board known as Blue Pill, LM016L LCD, ESP8266 board, and a few other components. The power requirement of this project is 5V for STM32 and 3.3V ~ 3.6V for ESP8266

Wi-Fi module.

5V is converted to 3.6V through the potential divider circuit formed by silicon diodes D3 and D4 and the 1k resistor connected in series with them. Each silicon diode's cut in voltage is 0.7V. So, the two diodes in series drop the voltage by 1.4V and we get $5V - 1.4V = 3.6V$.

As shown in Fig. 3, the DHT11 sensor is used to measure temperature and humidity. The sensor comes with a dedicated in-built NTC thermistor to measure temperature. It has an 8-bit microcontroller on-board to output the values of temperature and humidity as serial data through one-wire protocol. The sensor has only one data pin through which both temperature and humidity values can be read, thus saving pins on the microcontroller side. The sensor is factory calibrated and easily interfaces with a microcontroller. Pin details

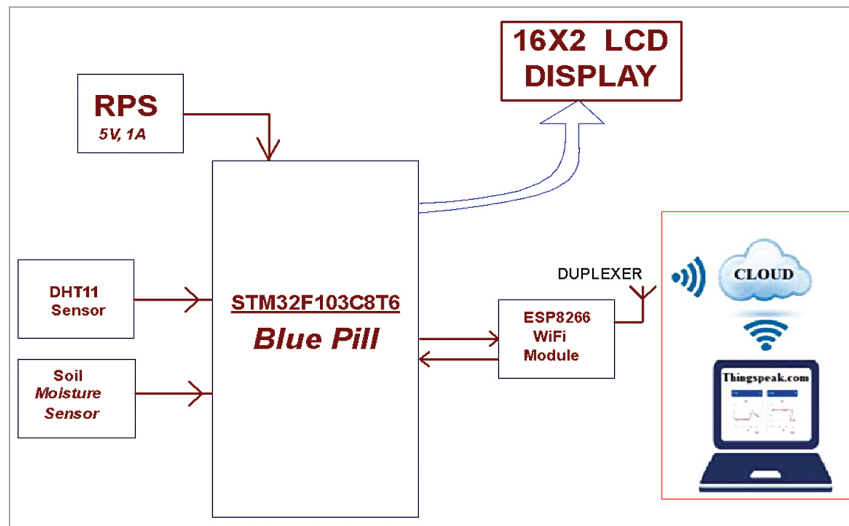


Fig. 1: Block diagram of the project

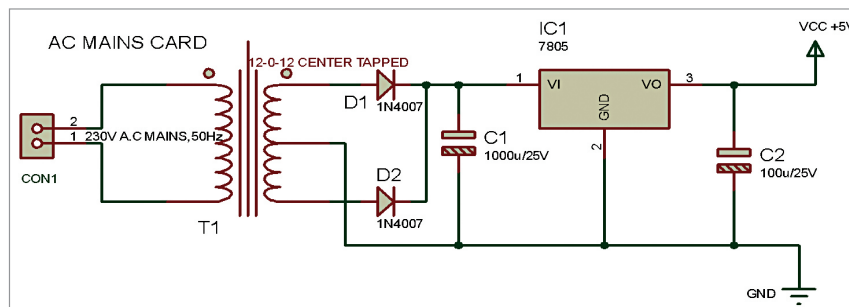


Fig. 2: Regulated power supply

PARTS LIST

Semiconductors:

- IC1 - 7805, 5V voltage regulator
- D1-D4 - 1N4007 rectifier diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon), unless stated otherwise:

- R1-R3 - 1-kilo-ohm

Capacitors:

- C1 - 1000 μ F, 25V electrolytic
- C2 - 100 μ F, 25V electrolytic

Miscellaneous:

- CON1 - 2-pin connector
- X1 - 230V AC primary to 12V-0-12V, 750mA secondary transformer
- LCD1 - LCDLM016L
- STM32F103 board (Blue Pill)
- ESP8266 board
- Soil moisture sensor
- DHT11 sensor
- Flexible wire